SignLens: AI-Based Sign Language Interpreter

Bridging the Communication Gap for the Deaf and Hard of Hearing

HACKNOVA 2025 • AI for Accessibility and Social Inclusion

Breaking Down Communication Barriers

The Challenge

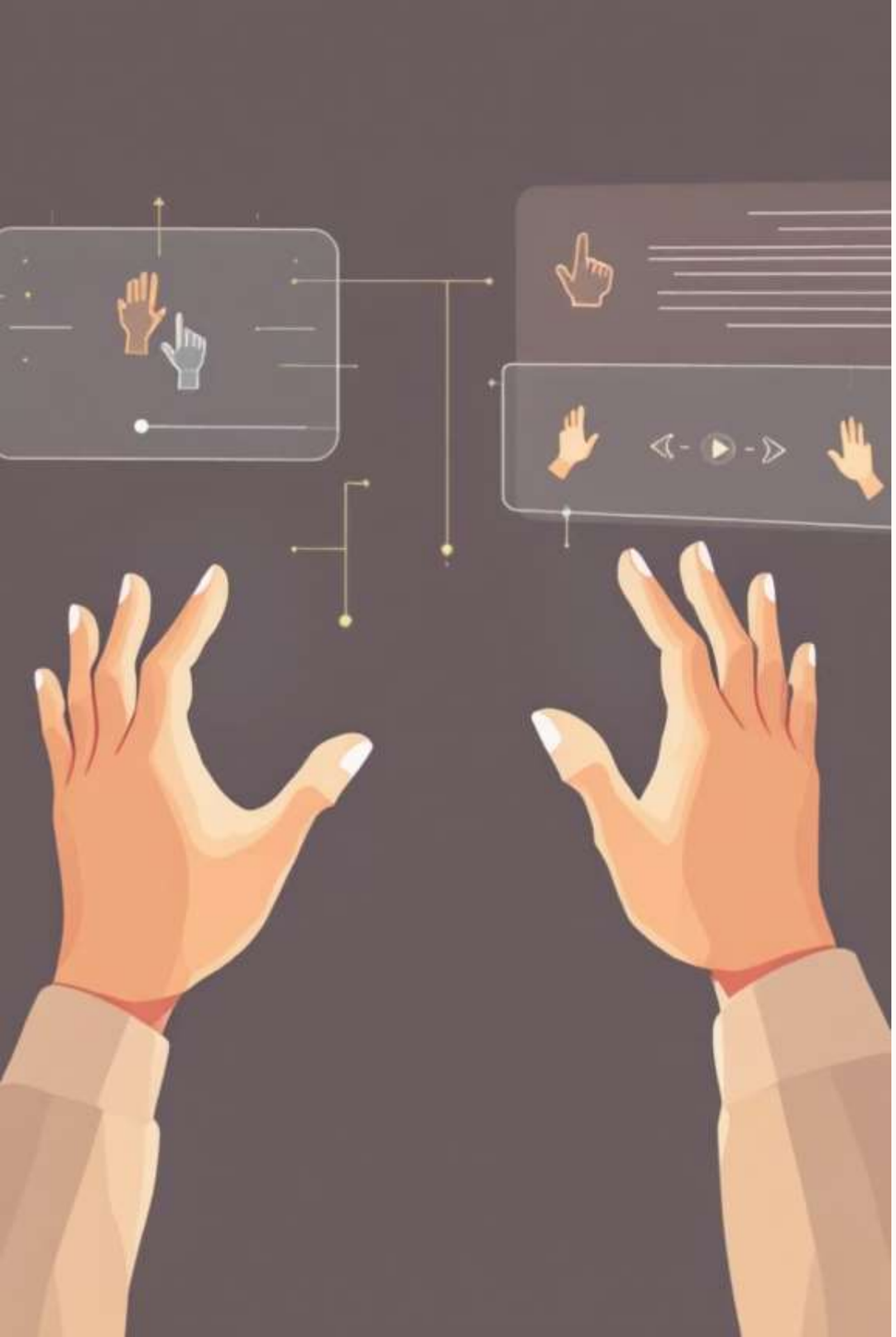
Over 466 million people worldwide are deaf or hard of hearing. Daily interactions—ordering at a café, attending a meeting, or simply chatting with family—become challenging when sign language users meet non-signers.

Current solutions are expensive, require specialized hardware, or aren't real-time. We need an accessible bridge.

Our Solution: SignLens

A real-time, web-based application that uses a standard webcam to capture sign language gestures and instantly translates them into on-screen text and spoken audio.

Two-way communication: It also captures spoken words via microphone and translates them back into text for sign language users.



How SignLens Works

Capture

Webcam captures sign language gestures in real-time

Track

MediaPipe extracts hand and body landmarks

Recognize

TensorFlow model predicts signed words

Translate

Output as text and spoken audio

No special hardware required—just a standard webcam, microphone, and internet connection. SignLens makes accessibility truly accessible.

Innovation That Sets Us Apart

Real-Time Translation

Instant conversion of gestures to text and speech with minimal latency, enabling natural conversation flow

Bidirectional Communication

Combines gesture recognition (Sign-to-Text) with speech recognition (Speech-to-Sign) for true dialogue

Web-Based & Lightweight

No app downloads or specialized equipment—works directly in any modern browser

Accessible by Design

Unlike expensive interpreter services or bulky hardware solutions, SignLens democratizes access to communication technology. It works on any device with a webcam and internet connection.

Built for Real Life

Perfect for everyday scenarios: shopping, classroom learning, medical appointments, workplace meetings, and family conversations at home.

Technical Architecture

Computer Vision

MediaPipe provides real-time hand, face, and body pose tracking with high precision

Machine Learning

TensorFlow powers our custom sign language classification model

Backend Server

Flask lightweight Python framework serves the model and handles API requests

Frontend & Speech

HTML/CSS/JavaScript with **Web Speech API** for browser-based audio processing

From Gesture to Speech: The Pipeline

01

Video Stream Input

Webcam continuously captures user's signing at 30+ frames per second

02

Landmark Extraction

MediaPipe identifies 21 hand landmarks plus face and body keypoints from each frame

03

Feature Processing

Landmark coordinates are normalized and prepared as input vectors for the model

04

Model Classification

TensorFlow neural network predicts the signed word or phrase with confidence scores

05

Output Generation

Predicted text displays on-screen while Web Speech API converts it to natural-sounding speech

Reverse Pipeline: For spoken input, the Web Speech API captures and transcribes audio, then displays it as text for the sign language user—enabling true two-way communication.

Feasibility and Implementation Strategy

High Feasibility

All core technologies—MediaPipe, TensorFlow, Flask, and Web Speech API—are mature, open-source, and extensively documented with active communities.

Hackathon Scope: A functional prototype focusing on 50-100 high-priority words and phrases is achievable within the competition timeframe.

Smart Development Approach

- Start with robust public datasets (WLASL, AUTSL, ISL datasets)
- Apply transfer learning from pre-trained gesture models
- Focus initially on Indian Sign Language (ISL) vocabulary

Challenges and Mitigation Strategies

Accuracy Variations

Challenge: Different signing styles, speeds, and co-articulation patterns

Strategy: Train on diverse datasets with multiple signers; implement confidence thresholds

Environmental Factors

Challenge: Poor lighting, difficult camera angles, background distractions

Strategy: Real-time environment quality checks; user guidance for optimal setup

Data Requirements

Challenge: Expanding vocabulary needs large, diverse training datasets

Strategy: Leverage existing datasets; implement user feedback loops; crowdsource contributions

We're also building a "Did you mean...?" feature that offers alternative translations and allows users to provide correction feedback, continuously improving accuracy over time.

Transformative Impact

Personal Empowerment

The deaf and hard-of-hearing community gains greater independence in daily communication—ordering food, asking for directions, or expressing needs without relying on interpreters.

Social Inclusion

Seamless interaction in public spaces (stores, banks, hospitals) and private settings (family gatherings) reduces isolation and strengthens relationships across hearing differences.

Economic Opportunity

Improved access to employment through better workplace communication, job interviews, and customer service interactions opens new career pathways.

Educational Access

Students in mainstream classrooms can participate more fully, engage with teachers and peers, and access learning opportunities without communication barriers.

Research Foundation

- **Google AI Blog (2020)**

  "MediaPipe Hands: On-device Real-time Hand Tracking" — Technical foundation for gesture recognition

- **Kaggle Datasets**

  ASL Alphabet & Indian Sign Language (ISL) Gestures — Training data for model development

- **Academic Research**

  "A Real-Time Sign Language Recognition System using Deep Learning and MediaPipe" — Methodology validation

- **TensorFlow Hub**

  Pre-trained models for gesture recognition and pose estimation — Transfer learning resources

- **MDN Web Docs**

  Web Speech API specification and browser compatibility — Implementation guidance

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SignLens: Making Communication Accessible for All

By combining cutting-edge AI with human-centered design, we're building more than an app—we're building bridges between communities, breaking down barriers, and creating a more inclusive world where everyone can be heard.